Supplemental Appendix S1.

Left Atrial Appendage Thrombus (LAAT) + Transesophageal Echocardiography (TEE) + Novel Antagonist Oral Anticoagulants (NOACs) Systematic Review

Search Methods Section
Prepared by: Lindsey Sikora, University of Ottawa
Date: September 12, 2018

Methods

The following databases were searched by a health sciences librarian (LS) during the electronic component of the systematic review: Medline and Medline in Process (via OVID), Embase Classic + Embase (via OVID), Cochrane's Central Registry for Randomized Controlled Trials, CENTRAL (via OVID), PsycINFO (via Ovid), and PubMed. A search strategy was developed in Medline, and then translated into the other databases, as appropriate (see Appendix 1). All databases were searched from the date of inception to September 11, 2018. There were no language exclusion criteria, nor any other publication restrictions. All references were entered into an Endnote file for processing (n = 2636).

Database searches (numbers for PRISMA flowchart):
Medline in Process and Medline (via OVID): n = 603
Embase (via OVID): n = 1346
CENTRAL (via OVID): n = 80
PubMed: n = 607

Total: n = 2636
Duplicates: n = 935

Total with duplicates removed: n = 1701

Medline and CENTRAL:
1. Anticoagulants/
2. Antithrombins/
3. ((novel or direct) adj2 (anticoagulant* or anti-coagulant*)).tw.
4. (noac* or doac*).tw.
5. (anticoagulant* or anti-coagulant* or antithrombin* or anti-thrombin*).tw.
6. (thrombin adj3 inhibit*).tw.
7. Factor Xa Inhibitors/
8. (Factor Xa adj2 (antagonist? or inhibit* or block*)).tw.
9. exp Vitamin K/ai
10. (Vitamin K adj2 (antagonist? or inhibit* or block*)).tw.
11. Dabigatran/

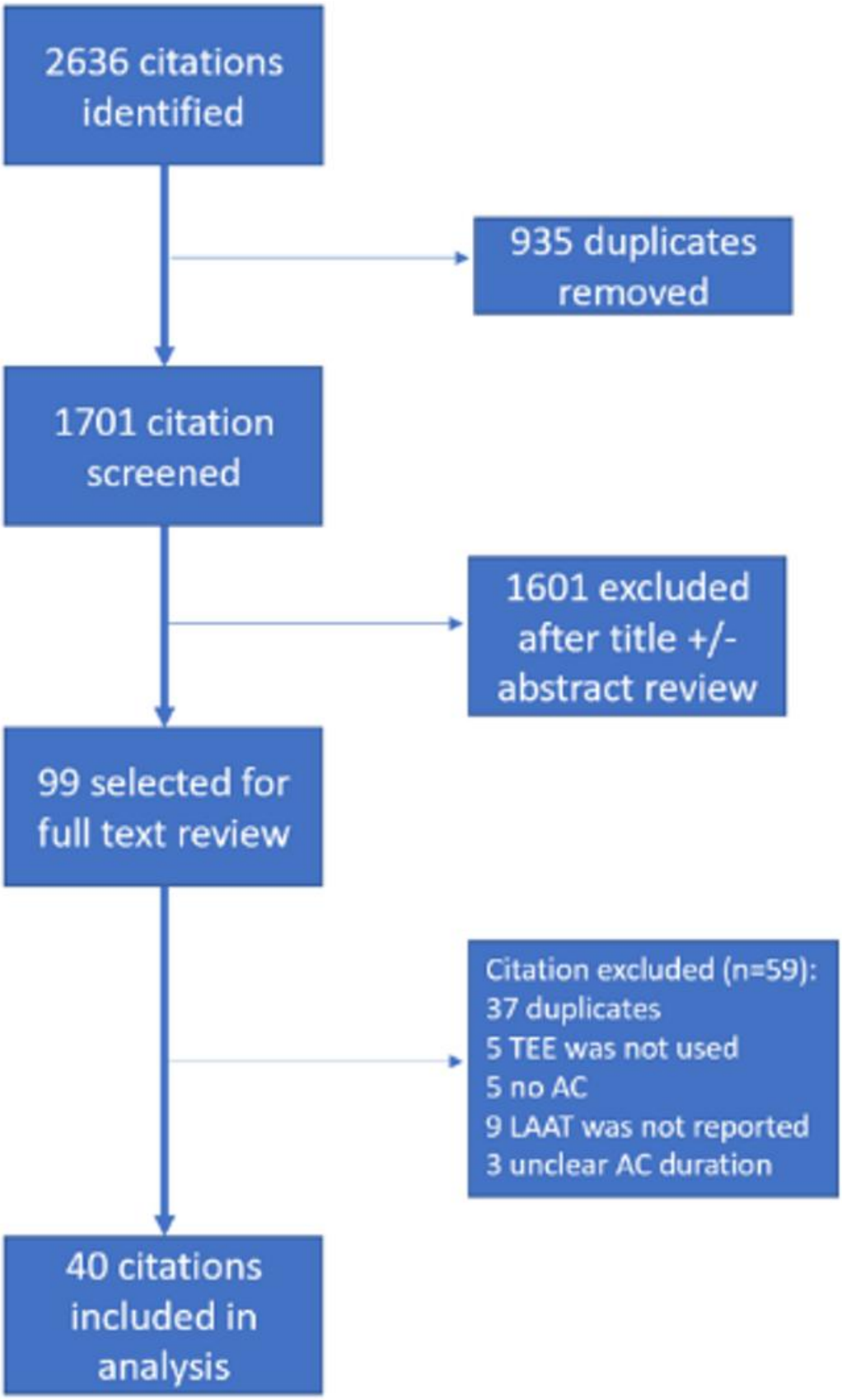
12. (dabigatran or pradaxa or bibr953 or bibr 953).tw.
13. Rivaroxaban/
14. (rivaroxaban or xarelto or bay597939 or bay 59 7939 or bay59 7939 or bay 597939).tw.
15. (apixaban or eliques or eliquis or bms562247 or bms 562247).tw.
16. (edoxaban or lixiana or du176 or du 176 or endoxaban or roteas or savaysa).tw.
17. or/1-16
18. Echocardiography, Transesophageal/
19. (transesophageal adj3 echocardiograph*).tw.
20. (transesophageal adj3 echocardiogram*).tw.
21. ((intracardiac or intra-cardiac) adj3 echo*).tw.
22. (tee or ice).tw.
23. or/18-22
24. atrial fibrillation/ or atrial flutter/
25. ((atrial or auricular) adj3 (fibrillation* or flutter* or sludg*)).tw.
26. spontaneous echocardiographic contrast*.tw.
27. SEC.tw.
28. cardioversion*.tw.
29. or/24-28
30. 17 and 23 and 29

Embase
1. Anticoagulants/
2. Antithrombins/
3. ((novel or direct) adj2 (anticoagulant* or anti-coagulant*)).tw.
4. (noac* or doac*).tw.
5. (anticoagulant* or anti-coagulant* or antithrombin* or anti-thrombin*).tw.
6. (thrombin adj3 inhibit*).tw.
7. Factor Xa Inhibitors/
8. (Factor Xa adj2 (antagonist? or inhibit* or block*)).tw.
9. antivitamin k/
10. (Vitamin K adj2 (antagonist? or inhibit* or block*)).tw.
11. Dabigatran/
12. (dabigatran or pradaxa or bibr953 or bibr 953).tw.
13. Rivaroxaban/
14. (rivaroxaban or xarelto or bay597939 or bay 59 7939 or bay59 7939 or bay 597939).tw.
15. (apixaban or eliques or eliquis or bms562247 or bms 562247).tw.
16. (edoxaban or lixiana or du176 or du 176 or endoxaban or roteas or savaysa).tw.
17. or/1-16
18. Echocardiography, Transesophageal/
19. (transesophageal adj3 echocardiograph*).tw.
20. (transesophageal adj3 echocardiogram*).tw.
21. intracardiac echocardiography/
22. ((intracardiac or intra-cardiac) adj3 echo*).tw.
23. (tee or ice).tw.
24. or/18-23
25. atrial fibrillation/ or atrial flutter/
26. ((atrial or auricular) adj3 (fibrillation* or flutter* or sludg*)).tw.
27. spontaneous echocardiographic contrast*.tw.
28. SEC.tw.
29. cardioversion*.tw.

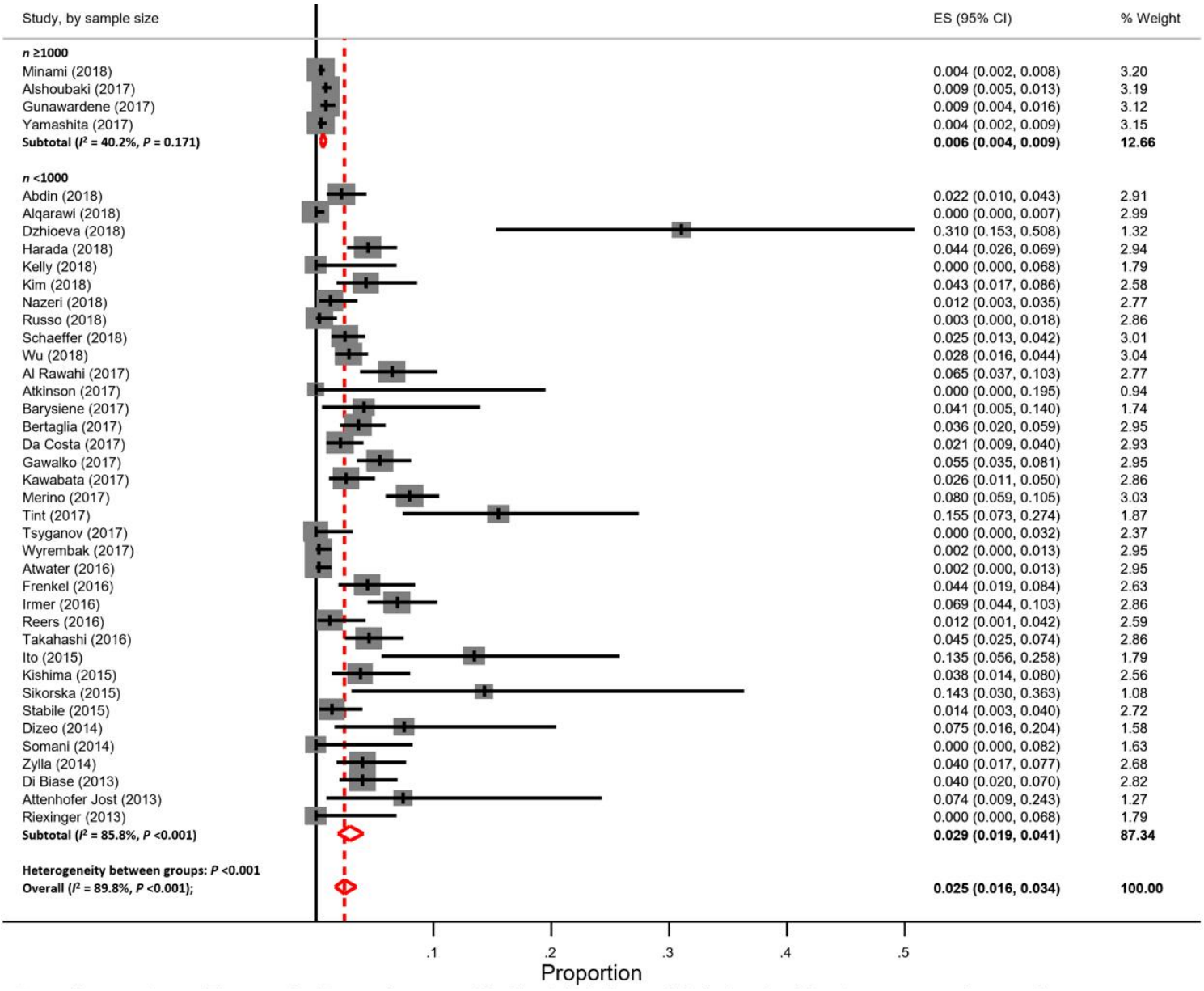
30. or/25-29
31. 17 and 24 and 30

PubMed

((((((((((atrial fibrillation[MeSH Terms]) OR atrial flutter[MeSH Terms]) OR atrial fibrillation*[Title/Abstract]) OR auricular fibrillation*[Title/Abstract]) OR atrial flutter*[Title/Abstract]) OR auricular flutter*[Title/Abstract]) OR atrial sludg*[Title/Abstract]) OR spontaneous echocardiographic contrast*[Title/Abstract]) OR (SEC[Title/Abstract] OR cardioversion*[Title/Abstract]))) AND ((((((echocardiography, transesophageal[MeSH Terms]) OR transesophageal echocardiograph*[Title/Abstract]) OR transesophageal echocardiogram*[Title/Abstract]) OR intracardiac echo*[Title/Abstract]) OR intra-cardiac echo*[Title/Abstract]) OR (TEE[Title/Abstract] OR ICE[Title/Abstract]))) AND ((((((((((((anticoagulants[MeSH Terms]) OR antithrombins[MeSH Terms]) OR novel anticoagulant*[Title/Abstract]) OR direct anticoagulant*[Title/Abstract]) OR (NOAC[Title/Abstract] OR DOAC[Title/Abstract])) OR (anticoagulant*[Title/Abstract] OR antithrombin*[Title/Abstract] OR anti-coagulant*[Title/Abstract] OR anti-thrombin*[Title/Abstract])) OR factor xa inhibitors[MeSH Terms]) OR factor xa inhibitor*[Title/Abstract]) OR Vitamin K antagonist[MeSH Terms]) OR Vitamin K antagonist*[Title/Abstract]) OR dabigatran[MeSH Terms]) OR (dabigatran[Title/Abstract] OR pradaxa[Title/Abstract] OR bibr953[Title/Abstract] OR bibr 953[Title/Abstract])) OR rivaroxaban[MeSH Terms]) OR (rivaroxaban[Title/Abstract] OR xarelto[Title/Abstract] OR bay597939[Title/Abstract] OR bay 59 7939[Title/Abstract] OR bay59 7939[Title/Abstract] OR bay 597939[Title/Abstract] OR apixaban[Title/Abstract] OR eliques[Title/Abstract] OR eliquis[Title/Abstract] OR bms562247[Title/Abstract] OR bms 562247[Title/Abstract] OR edoxaban[Title/Abstract] OR lixiana[Title/Abstract] OR du176[Title/Abstract] OR du 176[Title/Abstract] OR endoxaban[Title/Abstract] OR roteas[Title/Abstract] OR savaysa[Title/Abstract]))



Supplemental Figure S1. Literature search strategy



Supplemental Figure S2. Prevalence of left atrial thrombi detected transesophageal echocardiography in patients with atrial fibrillation on direct oral anticoagulants, stratified by sample size.

Supplemental Table S1: Characteristics of include studies

Study First Author	Year published	Publication type	CHADS2	CHAD-VASc	Paroxysmal AF (%)	Persistent (%)	Long standing (%)	Atrial flutter (%)	TEE strategy	Quality assessment
Pre AF ablation:										
Alqarawi[1]	2019	Manuscript	1±1 (mean)	1.9±1.4 (mean)	478 (72)	156 (24)	35 (5)	0 (0)	Routine	Low
Dzhioeva[2]	2018	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Harada[3]	2018	Manuscript	1.2±1.2 (mean)	2.1±1.6 (mean)	Not reported	144 (35)	Not reported	Not reported	Selective	Low
Wu[4]	2018	Manuscript	1 (0-2) (median)	2 (1-3) (median)	367 (60)	195 (32)	12 (2)	192 (32)	Selective	Low
Atkinson[5]	2017	Manuscript	1(0-2) (median)	0 (0-1) (median)	227 (68)	105 (32)	0 (0)	0 (0)	Selective	Low
DaCosta[6]	2017	Manuscript	Not reported	1.64 (1.46) (mean)	340 (59)	191 (33)	45 (8)	0 (0)	Selective	Low
Gunawardene[7]	2017	Manuscript	Not reported	Not reported	804 (48)	854 (52)	0 (0)	0 (0)	Routine	Low
Tsyganov[8]	2017	Manuscript	Not reported	2.09 (0.91) (mean)	71 (62)	38 (33)	6 (5)	0 (0)	Selective	Low
Wyrembak[9]	2017	Manuscript	Not reported	3.1±2 (mean)	385 (41)	478 (51)	74 (7.9)	0 (0)	Selective	Low

Yamashita[10]	2017	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Atwater[11]	2016	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Routine	Low
Frenkel[12]	2016	Manuscript	Not reported	2 (1-3.3) (median)	204 (53)	132 (34)	11 (3)	133 (34)	Selective	Very low
Sikorska[13]	2015	Manuscript	1.1 (1.07) (mean)	1.54 (1.38) (mean)	117 (73)	34 (21)	0 (0)	0 (0)	Selective	Low
Somani[14]	2014	Manuscript	0.74 (mean)	Not reported	140 (68)	67 (32)	0 (0)	0 (0)	Routine	Low
Riexinger[15]	2013	Abstract	Not reported	Not reported	Not reported	Not reported	13 (25)	Not reported	Routine	Low
Pre cardioversion:										
Kim[16]	2018	Manuscript	Not reported	1.7±1.3 (mean)	(specific type not reported): 327 (95)			17 (5)	Selective	Low
Russo[17]	2018	Manuscript	Not reported	Not reported	0(0)	510 (100)	0 (0)	0 (0)	Selective	Low
Schaeffer[18]	2018	Manuscript	Not reported	3 (0-9)	(specific type not reported): 924(72)			362(28)	Selective	Low
Barysienė[19]	2017	Manuscript	Not reported	3.5 (1.5) (mean)	0 (0)	437 (100)	0 (0)	0 (0)	Selective	Low
Merino[20]	2017	Abstract	Not reported	Not reported	270 (12)	909 (41)	0 (0)	0 (0)	Selective	Low

Tint[21]	2017	Manuscript	Not reported	Not reported	0 (0)	103 (100)	0 (0)	0 (0)	Routine	Low
Irmer[22]	2016	Abstract	Not reported	2.9 (1.7) (mean)	0 (0)	628 (100)	0 (0)	0 (0)	Routine	Low
Reers[23]	2016	Manuscript	Not reported	Not reported	172 (57%)	111 (37%)	2 (0.1%)	17 (6%)	Selective	Low
Stabile[24]	2015	Manuscript	Not reported	2.37 (1.44) (mean)	Not reported	Not reported	Not reported	Not reported	Selective	Low
Dizeo[25]	2014	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Routine	Low
Mixed										
Di Biase[26]	2013	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Kelly[27]	2018	Abstract	Not reported	4 (mean)	Not reported	Not reported	Not reported	Not reported	Selective	Low
Minami[28]	2018	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Al Rawahi[29]	2017	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Alshoubaki[30]	2017	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Routine	Low
Bertaglia[31]	2017	Manuscript	Not reported	2.6 (1.2) (median)	53 (13)	361 (87)	0 (0)	0 (0)	Selective	Low

Gawalko[32]	2017	Manuscript	2 (2-3) (median)	2 (1-3) (median)	492 (57)	310 (36)	57 (7)	0 (0)	Routine	Low
Kawabata[33]	2017	Manuscript	1.2 (1.1) (mean)	1.9 (1.5) (mean)	284 (51)	275 (49)	0 (0)	0 (0)	Selective	Low
Ito[34]	2015	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Routine	Low
Kishima[35]	2015	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Zylla[36]	2014	Manuscript	Not reported	4 (median)	210 (33)	153 (24)	113 (18)	53 (8)	Selective	Very low
Attenhofer Jost[37]	2013	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Not reported										
Abdin[38]	2018	Abstract	Not reported	4 (median)	Not reported	Not reported	Not reported	Not reported	Selective	Low
Nazeri[39]	2018	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low
Takahashi[40]	2016	Abstract	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Selective	Low

TEE: trans-esophageal echocardiography

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